

XIN NA (纳鑫)

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EDUCATION

Tsinghua University, School of Software, Ph.D. Student 2020.8 - Present

- Advised by Prof. [Yuan He](#).

Tsinghua University, School of Software, Bachelor of Engineering 2016.8 - 2020.7

- With honor: **Excellent Graduate** in Tsinghua University and **Outstanding Graduate** in School of Software.

RESEARCH INTERESTS

My research interests lie in wireless networking and ubiquitous IoT, especially battery-free and low-power IoT, cross-technology communication, and wireless sensing. I strive to develop high-performance and reliable IoT systems that enable continuous, ubiquitous sensing across diverse environments. To achieve this vision of sensing *anywhere, anytime*, I focus on battery-free and low-power solutions, which I believe hold the greatest promise for scalable and sustainable deployment. My current research in pursuit of this goal includes:

1. Analog backscatter modulation of sensed data without microcontrollers.
2. Parallel RFID communication in the frequency domain via frequency-selective antennas.
3. Truly ubiquitous battery-free backscatter IoT leveraging ambient cellular signals.

PUBLICATIONS

[1] Analog Backscatter for Commodity WiFi with Payload Transparency

Xin Na, Yuan He, Xiuzhen Guo, Jia Zhang, Yang Zou, Zihao Yu, Yunhao Liu.

IEEE/ACM Transactions on Networking (ToN), 2025. [\[PDF\]](#)

[2] QuinID: Enabling FDMA-Based Fully Parallel RFID with Frequency-Selective Antenna

Xin Na, Jia Zhang, Jiacheng Zhang, Xiuzhen Guo, Yang Zou, Meng Jin, Yimiao Sun, Yunhao Liu, Yuan He.

ACM International Conference on Mobile Computing and Networking (MobiCom), 2025. [\[PDF\]](#) [\[Code\]](#)

[3] Leggiero: Analog WiFi Backscatter with Payload Transparency

Xin Na, Xiuzhen Guo, Zihao Yu, Jia Zhang, Yuan He, Yunhao Liu.

ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2023. [\[PDF\]](#) [\[Code\]](#)

[\[Slides\]](#)

[4] Wi-attack: Cross-technology Impersonation Attack against iBeacon Services

Xin Na, Xiuzhen Guo, Yuan He, Rui Xi.

IEEE International Conference on Sensing, Communication, and Networking (SECON), 2021. [\[PDF\]](#) [\[Code\]](#) [\[Slides\]](#)

[5] Satori: In-band Analog Backscatter for Audio Transmission

Yang Zou, **Xin Na**, Yimiao Sun, Yande Chen, Yuan He.

ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025. [\[PDF\]](#) (**Best Paper Nominee, 2 out of 233.**)

[6] SmarTiSCH: An Interference-Aware Engine for IEEE 802.15.4e-based Networks

Zihao Yu, **Xin Na**, Carlo Alberto Boano, Yuan He*, Xiuzhen Guo, Pengyu Li, Meng Jin.

ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), 2022. [\[PDF\]](#)

[7] Trident: Interference Avoidance in Multi-reader Backscatter Network via Frequency-space Division

Yang Zou, **Xin Na**, Xiuzhen Guo, Yimiao Sun, Yuan He.

IEEE International Conference on Computer Communications (INFOCOM), 2024. [\[PDF\]](#)

[8] mmHawkeye: Passive UAV Detection with a COTS mmWave Radar

Jia Zhang, **Xin Na**, Rui Xi, Yimiao Sun, Yuan He.

IEEE International Conference on Sensing, Communication, and Networking (SECON), 2023. [\[PDF\]](#)

[9] Trident: Interference Avoidance in Multi-reader Backscatter Network via Frequency-space Division

Yang Zou, **Xin Na**, Yimiao Sun, Yuan He.

IEEE/ACM Transactions on Networking (ToN), 2024. [\[PDF\]](#)

- [10] BIFROST: Reinventing WiFi Signals Based on Dispersion Effect for Accurate Indoor Localization
Yimiao Sun, Yuan He, Jiacheng Zhang, **Xin Na**, Yande Chen, Weiguo Wang, Xiuzhen Guo.
ACM Conference on Embedded Networked Sensor Systems (SenSys), 2023. [PDF]
- [11] Detection and Identification of Non-cooperative UAV Using a COTS mmWave Radar
Yuan He, Jia Zhang, Rui Xi, **Xin Na**, Yimiao Sun, Beibei Li.
ACM Transactions on Sensor Networks (TOSN), 2024.[PDF]
- [12] RFinder: Pinpoint the Invisible RFID Tags in the Prefabricated Buildings
Jiacheng Zhang, Meng Jin, Yimiao Sun, Weiguo Wang, Jia Zhang, **Xin Na**, Xiuzhen Guo, Yuan He.
International Conference on Embedded Wireless Systems and Networks (EWSN), 2024.[PDF]
- [13] Taming the Errors in Cross-Technology Communication: A Probabilistic Approach
Xiuzhen Guo, Yuan He, Jia Zhang, Haotian Jiang, Zihao Yu, **Xin Na**.
ACM Transactions on Sensor Networks (TOSN), 2021.[PDF]
- [14] Real-Time Continuous Activity Recognition With a Commercial mmWave Radar
Yunhao Liu, Jia Zhang, Yande Chen, Weiguo Wang, Songzhou Yang, **Xin Na**, Yimiao Sun, Yuan He.
IEEE Transactions on Mobile Computing (TMC), 2024.[PDF]
- [15] Cross-Technology Communication for the Internet of Things: A Survey
Yuan He, Xiuzhen Guo, Xiaolong Zheng, Zihao Yu, Jia Zhang, Haotian Jiang, **Xin Na**, Jiacheng Zhang.
ACM Computing Surveys (CSUR), 2022.[PDF]
- [16] A Survey of mmWave-Based Human Sensing: Technology, Platforms and Applications
Jia Zhang, Rui Xi, Yuan He, Yimiao Sun, Xiuzhen Guo, Weiguo Wang, **Xin Na**, Yunhao Liu, Zhenguo Shi, Tao Gu.
IEEE Communications Surveys & Tutorials (COMST), 2023.[PDF]

TOOLS AND OPEN-SOURCE RESEARCHES

Beyond research, I also dedicate myself to building and sharing tools, hardware, and software to advance the wireless networking community.

QuinID, hardware and software for an FDMA-based fully parallel RFID system.

- The UHF RFID band is divided into five sub-bands, resulting in a fivefold improvement in reading rate.
- An additional tool, **UHF-Gen2-RFID-Reader**, is provided—a high-performance SDR-based reader for the broad RFID research community, achieving up to 1000 reads per second.

Leggiero, the entire hardware of an analog WiFi backscatter.

- The tag converts the analog sensor voltage to the RF signal phase and embeds it in the CSI.
- An updated version, Leggiero+, is also provided with a few enhancements made to the circuit design.

Wi-attack, the software of a cross-technology impersonation attack method against iBeacon.

- Uses WiFi devices to attack iBeacon services by impersonating an iBeacon base station.

PROFESSIONAL ACTIVITIES

ACM Transactions on Internet of Things, Reviewer	2025
IEEE Transactions on Mobile Computing, Reviewer	2024 - Now
Contributed to paper reviews for INFOCOM, IPSN, SenSys, SECON	2021 - Now
Teaching Assistant, Network Measurement and Analysis, Instructor: Prof. Yuan He	2021 - 2024

HONORS AND AWARDS

Talents Scholarship (<i>Top 15%</i>), Tsinghua University	2021 - 2024
Excellent University Graduate (<i>Top 10%</i>), Tsinghua University	2020.6
Outstanding Graduate (<i>Top 10%</i>), School of Software	2020.6
National Encouragement Scholarship (<i>Top 5%</i>), Tsinghua University	2018, 2019